

Volumetric Fan-Beam Imaging with Neural Attenuation Field for Cone-Beam Artifact Reduction in CBCT

Abstract

Background: Although cone-beam computed tomography (CBCT) acquires a complete volumetric dataset in a single gantry rotation, a standard circular trajectory is inherently affected by cone-beam artifacts because the acquired data does not satisfy the Tuy data sufficiency condition. Off-plane structures are incompletely sampled in Radon space, leading to axial blurring, intensity distortion, and structure loss that become more pronounced at larger cone angles.

Purpose: This study aims to reduce cone-beam artifacts in CBCT with an unsupervised neural attenuation field (NAF) reconstruction pipeline, which decouples the reconstruction geometry from the imaging acquisition geometry and requires no external training data.

Methods: We propose Volumetric Fan-beam Imaging with Neural Attenuation Field (VFINAF) for cone beam CT. In this method, the NAF is first trained to fit the measured cone-beam projection data. The trained NAF is then used for dense virtual fan-beam re-projection for each axial plane, producing individual slice fan-beam sinograms for 2D CT image reconstruction using conventional algorithms, such as FBP and SIRT. Volumetric 3D CT images are assembled from these 2D CT slices. Because each 2D slice reconstruction is performed from complete two-dimensional fan-beam sinogram, it avoids the Tuy's incompleteness associated with the original cone-beam acquisition geometry. Initial evaluation of the VFINAF was done using a digital FORBILD-Defrise phantom. It was further tested using experimental CBCT data with a homemade Defrise phantom and the RANDO phantom. The images are compared to those

obtained by standard FDK and SIRT methods using visual assessment and quantitative measures including peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), line profiles.

Results: In the digital FORBILD-Defrise phantom, FDK preserved only structures within an approximately 5° cone-angle range and SIRT extended this to about 9° , whereas VFINAF recovered all eleven disks across the full 20° range with clear delineation of fine structures. VFINAF achieved the highest PSNR and SSIM at every evaluated cone angle, with SSIM remaining above 0.80 at the peripheral disks. In the physical Defrise phantom imaging, VFINAF resolved all twelve disks, compared with five for SIRT. In the RANDO head phantom imaging, VFINAF reduced shading, streaking, and loss of structural definition across the axial, sagittal, and coronal planes in comparison with conventional reconstruction, and showing closer visual agreement with the reference image acquired from the MDCT.

Conclusions: VFINAF significantly reduces cone-beam artifacts in CBCT. The virtual dense fan-beam re-projection by using the NAF continuity prior decoupled image reconstruction geometry from the original cone-beam acquisition geometry. This extends the effective FOV of CBCT while preserving the advantages of NAF-based reconstruction.

Key Words:

Cone-beam computed tomography; Cone-beam artifact reduction; Neural attenuation field; Field-of-view extension

1. Introduction

Cone-beam computed tomography (CBCT) is widely used for diagnostic evaluation, image guidance in radiation therapy, and interventional procedures¹⁻³. In contrast to multidetector helical CT, CBCT uses a flat-panel detector to acquire a series of two-dimensional projections during a single gantry rotation, covering a three-dimensional field of view that is subsequently reconstructed into a volumetric image.^{4,5} This acquisition configuration enables volumetric coverage within a single rotation and supports compact system designs⁶⁻⁹. However, the same cone-beam geometry also introduces an intrinsic reconstruction limitation, especially when the axial coverage or cone angle increases. Cone-beam artifacts arise from data incompleteness associated with Tuy's data sufficiency¹⁰. Only the central plane is sampled more completely, whereas off-plane structures miss information in Radon space. As a result, conventional CBCT reconstructions show axial blurring, intensity distortion, structure loss, and degraded image quality near the longitudinal edges of the volume. These artifacts are not only visual imperfections, but they can also reduce the reliability of anatomical interpretation, registration, segmentation, and quantitative image analysis. The problem becomes more pronounced as the cone angle increases because the acquired data fail to satisfy the completeness conditions required for voxel-based and algebraic and iterative reconstruction¹¹⁻¹⁴. Several approaches have been proposed to reduce cone-beam artifacts. Classical methods include projection weighting strategies (e.g., Parker-FDK¹⁵ and CW-FDK¹⁶), ray rearrangement methods (e.g., P-FDK¹⁷ and T-FDK¹⁸), and compensation terms added to FDK-type algorithms^{13,19,20}. These methods can reduce some cone-beam artifacts, but their performance remains limited at larger cone angles, where data incompleteness becomes more pronounced. More recently, supervised deep-learning approaches have been proposed for cone-beam artifact reduction. Han et al.²¹ developed a

differentiated-backprojection-domain convolutional neural network (CNN) that exploits the angular symmetry of CBCT, whereas Minnema et al.²² developed a geometry-aware mixed-scale dense network operating on cylindrical slices. These studies show that learned models can suppress cone-beam artifacts, but they generally correct images within the original cone-beam geometry and require paired or carefully matched training data.

Neural attenuation fields (NAFs), a medical-imaging extension of implicit neural representations (INRs), have shown promise for tomographic reconstruction²³⁻²⁶, particularly for sparse-view imaging²⁷⁻³¹ and metal artifact reduction³²⁻³⁶. Rather than storing the image on a discrete fixed voxel grid, a NAF represents the CT images as a continuous function field of spatial coordinates and optimizes this function by matching the measured projections, without requiring external training data. The inherent continuity of the INR provides a useful prior for sparse-view reconstruction, in which limited angular sampling can make reliable CT image reconstruction challenging³⁰. In early object-domain approaches, the trained NAF was used as the reconstructed image, even though the loss was evaluated only between the predicted and measured projections. Consequently, voxels that are weakly constrained by the sparse measurements may be governed more by the fitted network than by the data, and overfitting to the sparse-view sinogram can appear as noise-like intensity errors in the directly queried volume. CoIL instead represents the sinogram as a continuous function of measurement coordinates and synthesizes denser measurements for subsequent reconstruction²⁷. However, the view-angle and detector coordinates of a sinogram are coupled by the projection geometry rather than forming independent orthogonal spatial axes, which is inconsistent with the coordinate structure assumed by Fourier spatial encoding. SCOPE addresses these limitations by combining re-projection with

hash encoding of the spatial coordinates³⁰. Instead of using directly queried voxel values as the final image, it utilizes the learned NAF for forward projection to synthesize a dense-view sinogram in the same measurement space as the acquired data. The final image is then reconstructed using FBP, thereby reducing the voxel-wise overfitting associated with direct readout.

Cone-beam artifacts can also be viewed as a consequence of incomplete projection sampling^{13,14,21}, but the incompleteness is geometric rather than angular: sparse-view artifacts arise from insufficient angular sampling, whereas cone-beam artifacts arise from Tuy's incompleteness in off-plane regions. The continuity prior of a neural attenuation field helps compensate for the missing off-plane information caused by Tuy's incompleteness. However, existing NAF-based CBCT methods still use the original cone-beam geometry to synthesize additional projections. Consequently, the synthesized projections remain geometrically incomplete, leaving cone-beam artifacts largely unresolved^{26,27,30}.

Here we propose Volumetric Fan-beam with Imaging Neural Attenuation Field (VFINAF), an unsupervised reconstruction pipeline that decouples the reconstruction geometry from the acquisition geometry to reduce the cone beam artifacts. VFINAF first trains a NAF model from cone-beam projections. The trained NAF is then queried along dense virtual fan-beam rays to synthesize complete fan-beam sinograms slice-by-slice, which are reconstructed with conventional reconstructions, such as 2D FBP or SIRT. The advantages of VFINAF are three folds. First, the intrinsic continuity of NAF compensates the incompleteness problems. Second, dense-reprojection of NAF allows query a voxel from more directions compared with the

original acquisition geometry. Third, because each slice is reconstructed from a complete 2D fan-beam sinogram, the final reconstruction is not subject to off-plane limitations associated with Tuy's incompleteness. We hypothesize that this geometry-decoupled re-projection preserves some off-plane structures estimated by the NAF continuity prior, reducing cone-beam artifacts while preserving the advantages of NAF-based reconstruction.

2. Materials and Methods

2.A VFANAF Model

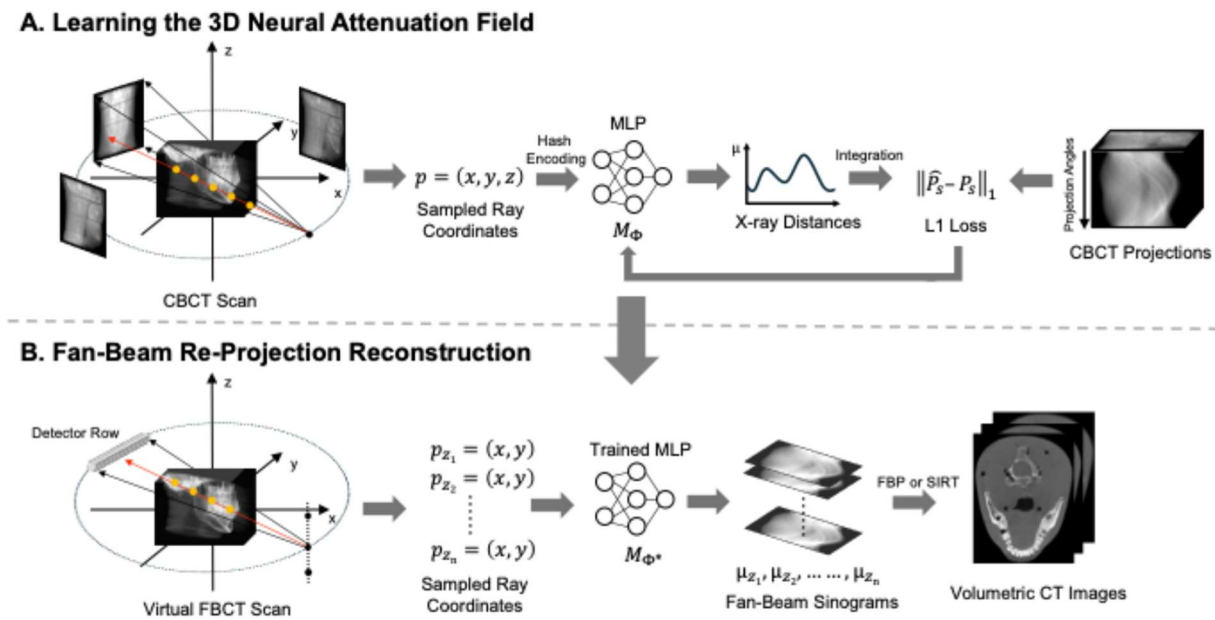


Figure 1. The VFANAF pipeline. (A) Learning the 3D neural attenuation field (NAF) from the measured cone-beam scan. Coordinates sampled along each X-ray are hash-encoded and passed to a multilayer perceptron (MLP, M_Φ) that returns the linear attenuation coefficient μ , which is integrated along the ray and compared with the measured projection through an L1 loss. (B) Fan-beam re-projection reconstruction. The trained MLP is queried slice by slice along virtual fan-beam rays to generate dense fan-beam sinograms, which are reconstructed by FBP or SIRT into the slice CT images. Volumetric CT images are assembled from these slice CT images.

VFANAF represents the scanned object as a continuous neural attenuation field (NAF) and recovers it by fitting the NAF to the measured projections. The NAF is a coordinate-based function

$$\mu(\mathbf{x}) = M_{\phi}(\mathbf{x}), \quad (1)$$

where $\mathbf{x} = (x, y, z)$ is a spatial coordinate, μ is the linear attenuation coefficient at that point, and Φ represents the trainable parameters of the MLP. Each measured projection is modeled as the line integral of μ along the corresponding X-ray path. Denoting the forward operator by a system matrix A , the predicted projections $\hat{P}$ are

$$\hat{P} = A \mu(\mathbf{x}) = A M_{\phi}(\mathbf{x}), \quad (2)$$

and the MLP parameters are learned by minimizing the discrepancy between the predicted projections $\hat{P}$ and the measured projections P ,

$$\Phi^* = \arg \min_{\Phi} \mathcal{L}_1(\hat{P}, P). \quad (3)$$

At each iteration, a batch of rays is drawn at random from the measured cone-beam rays, and the NAF is fit by minimizing an L_1 loss between the predicted and measured projections,

$$\mathcal{L}_1 = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} |\hat{P}(r) - P(r)|, \quad (4)$$

with $\mathcal{R}$ the ray batch. The training stops after certain number of iterations.

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The VFINAF pipeline is illustrated in Figure 1 and consists of two stages. In the first stage, Fig 1(a), the NAF is learned from the measured cone-beam projections. To enhance the recovery of high spatial frequency information, each spatial coordinate is first mapped to a feature vector $\mathbf{v}$ by a multi-resolution hash encoding³⁷, and the feature vector is decoded by a MLP into the attenuation coefficient,

$$\mathbf{v} = H_{\psi}(\mathbf{x}), \quad \mu = M_{\phi}(\mathbf{v}), \quad (5)$$

where ψ and ϕ are the parameters of the hash encoding and the MLP, and $\Phi = \{\psi, \phi\}$. The hash encoding maintains trainable feature grids at L resolution levels from coarse to fine, with each level stored in a fixed-size hash table. The cone-beam forward projection follows the Beer-

Lambert line-integral model. Along an X-ray, the NAF is sampled and integrated to give the predicted projection, collecting all cone-beam rays gives the compact form $\hat{P}_c = A_c M_\phi$.

After training, M_ϕ^* provides a continuous attenuation estimate throughout the volume, including off-plane regions affected by Tuy's incompleteness. In the second stage, Fig 1(b), the learned NAF is used to synthesize virtual projections, but in fan-beam rather than cone-beam geometry. For each axial slice $z = z_k$, a dense set of virtual fan-beam rays lying entirely within that plane is defined. Querying the NAF along these rays and integrating yields the fan-beam sinogram of the slice, $\hat{P}_f = A_f M_\phi$. VFINAF synthesizes significant denser view sinograms than raw acquire projection data. The volumetric image is then reconstructed slice by slice. Re-synthesizing in fan-beam geometry therefore separates the downstream reconstruction from the original cone-beam acquisition geometry and avoids the cone-beam reconstruction error of direct cone-beam image formation.

The architecture of NAF we adopted is similar to previous studies³⁸. The MLP has two hidden layers of 128 neurons each. The hash encoding uses $L = 16$ levels with table size $T = 2^{21}$, $F = 8$ features per entry, minimum resolution $N_{\min} = 2$, and per-level growth factor $b = 1.6$. The network is trained with the Adam optimizer at an initial learning rate of 10^{-3} , halved every 4000 epochs over a total of 12000 epochs. All experiments were run on a single NVIDIA RTX 5090 GPU on a workstation with 64 GB of system memory.

2.B Simulation Experiment

A FORBILD-Defrise digital phantom was generated as the simulation ground truth and was designed to expose cone beam artifacts along the vertical direction (Figure 2). Each disk slice

follows the FORBILD pattern³⁹; the disk thickness is 5 mm, the inter-disk air gap is 5 mm, and the phantom contains 11 disks in total along the axial direction. A relatively large cone angle of 20 degrees was chosen to mimic the maximum cone angle encountered in clinical CBCT. The source-to-object distance (SOD) is 298 mm and the source-to-detector distance (SDD) is 567 mm. In the simulation experiment the 360 raw projection images were generated over 360°, the 1440 views dense re-projection from NAF were used for FBP reconstruction, and the volumetric CT images were reconstructed at an isotropic voxel size of 0.5 mm.

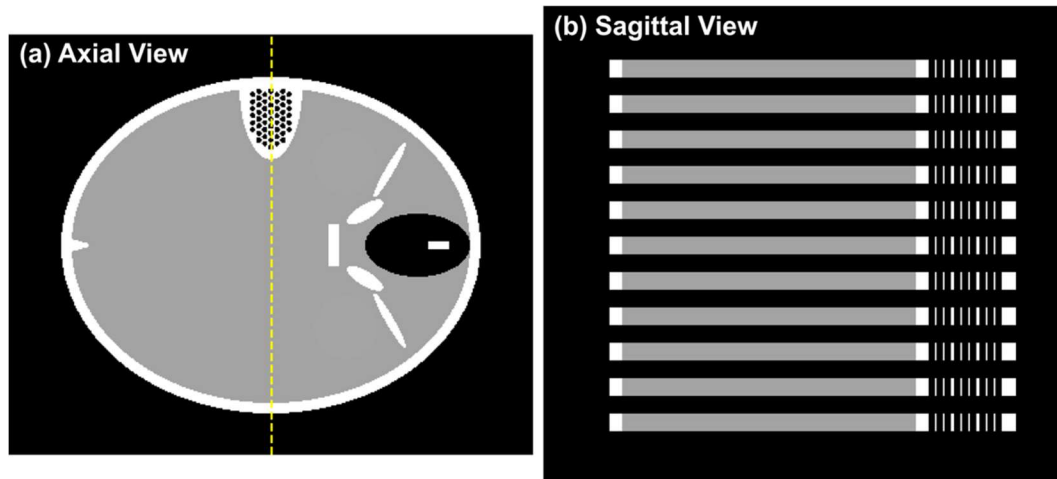


Figure 2. The FORBILD-Defrise digital phantom used in the simulation experiment: (a) axial (b) sagittal view. The phantom comprises eleven axially stacked FORBILD disks, each 5 mm thick and separated by 5 mm gaps.

2.C Real Experiment

Two physical phantoms were imaged on a CNT X-ray CBCT system⁴⁰ for real-data validation (Figure 3). The first is a homemade Defrise phantom, with a stack of 12 acrylic discs 160 mm in diameter and 4 mm thick. The discs are separated by 4 mm air gaps (Fig. 3a), which reproduces the off-plane disk structure of the digital FORBILD-Defrise phantom and is designed to investigate the cone-beam artifacts along the axial direction. The second is an anthropomorphic adult skull and tissue-equivalent head phantom (RANDO, Nuclear Associates, Hicksville, NY, Fig. 3b), which provides realistic craniofacial bony anatomy. The system geometry has SOD =

410 mm, SDD = 620 mm, and an x-ray beam cone angle of 10.4° . Projections were acquired with a flat-panel detector (Xineos-1511, Teledyne Dalsa, Waterloo, Canada) with an active area of 147.1 mm (width) by 113.7 mm (height) and a pixel pitch of 99 microns. The detector was shifted laterally by 70 mm, extending the field of view (FOV) to 187 mm (width) by 70 mm (height) at the rotation center⁷, and was operated in 2×2 binning mode. The CNT X-ray source (NuRay Technologies, Changzhou, China) operated at 110 kV with 1.7 mm Al inherent filtration and an additional 0.3 mm Cu external filter⁴¹. The per-source exposure time was 5 ms with a tube current of 11 mA, controlled by an electronic control module (ECS, NuRay Technologies). The total exposure was 19.8 mAs, corresponding to a dose-area product of $5.17 \text{ dGy} \cdot \text{cm}^2$. For anatomical reference, the RANDO head phantom was also scanned on a clinical multidetector CT system (SOMATOM Force, Siemens, Germany) at 110 kV, using the standard adult head protocol in our institution. The MDCT volume was reconstructed at a voxel size of $0.5 \times 0.5 \times 1 \text{ mm}$ and serves as the anatomical reference.

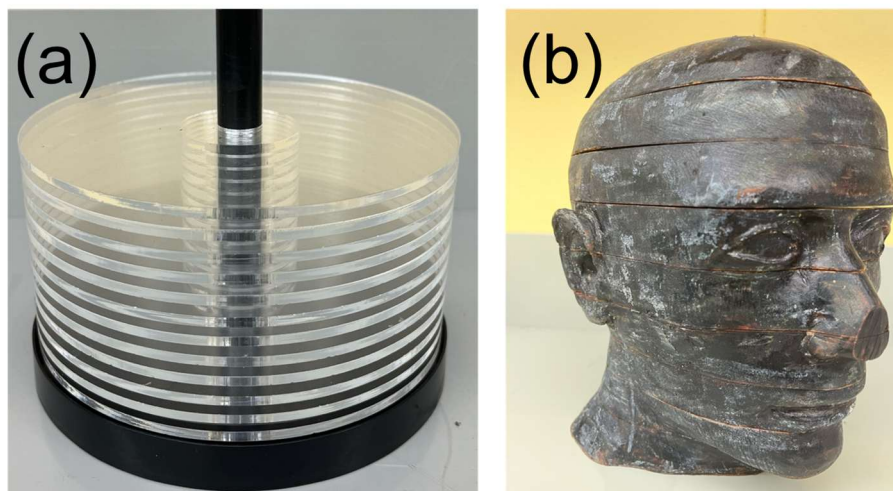


Figure 3. Physical phantoms used in this study. (a) A homemade Defrise phantom with a stack of acrylic discs. (b) An anthropomorphic adult head phantom (RANDO, Nuclear Associates, Hicksville, NY).

The 360 projections were acquired in cone-beam half-detector geometry over 360° , while VFANAF synthesizes 1440 dense projections in fan-beam full-detector geometry in re-projection

process. In the half-detector geometry, SIRT reconstruction was used as the downstream reconstruction for all physical phantom volumes. All datasets acquired from the physical phantom were reconstructed at an isotropic voxel size of 0.4 mm.

2.D Evaluation Metrics

Image quality was evaluated by Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM)^{42,43}. PSNR is defined as

$$\text{PSNR} = 10 \log_{10}(\text{MAX}^2/\text{MSE}) \quad (6)$$

where MAX is the maximum attainable intensity value of the reconstructed volume and MSE is the voxel-wise mean squared error against the ground truth.

SSIM is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (7)$$

with means μ_x and μ_y , variances σ_x^2 and σ_y^2 , covariance σ_{xy} , and stabilization constants C_1 and C_2 .

Throughout the Results section, for the digital phantom, the FORBILD-Defrise volume served as the ground truth. For the RANDO phantom, the MDCT image was used as the reference.

3. Results

3.A Simulation Results

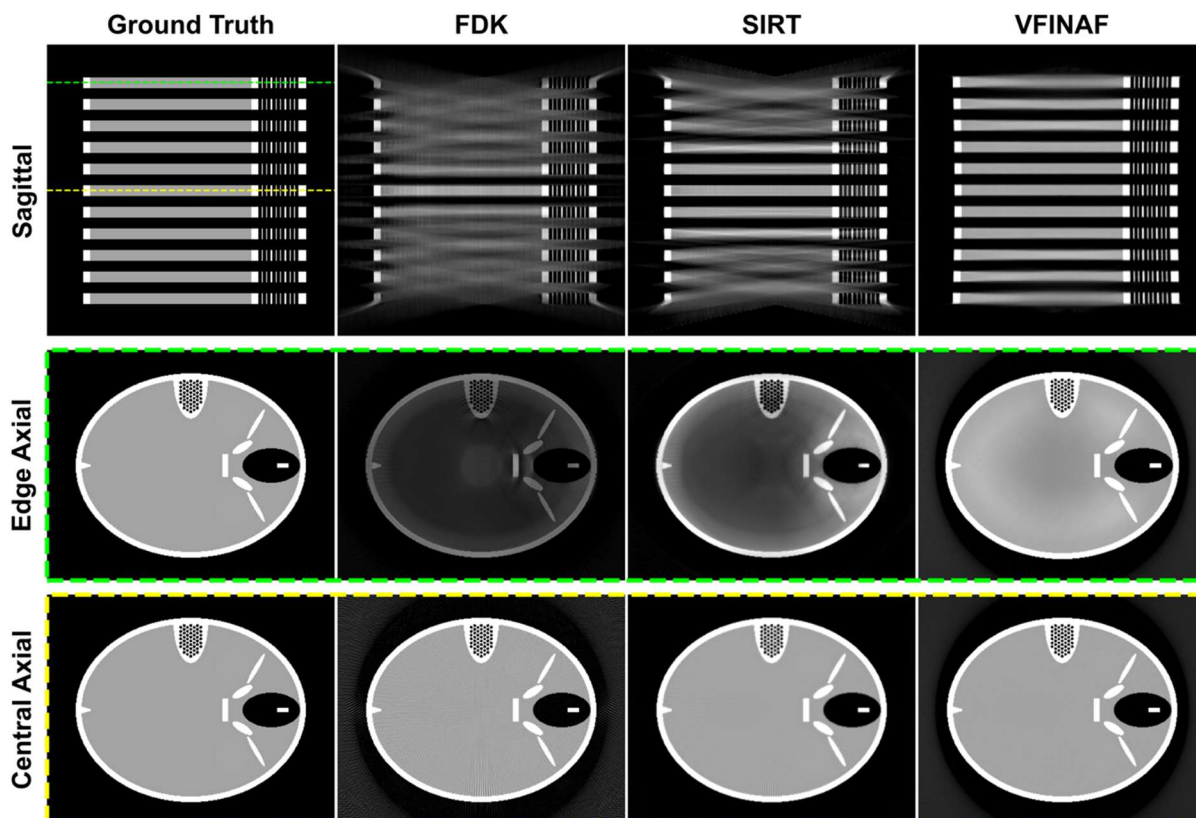


Figure 4. Cone-beam artifact reduction demonstration using the FORBILD-Defrise phantom. The columns show the ground truth and reconstructed images obtained with FDK, SIRT, and VFINAF. The rows show sagittal views, edge axial slices, and central axial slices. The dashed green and yellow lines in the sagittal view indicate the locations of the corresponding axial slices, which are outlined with matching dashed boxes. Window: $[0, 0.3] \text{ cm}^{-1}$.

The FORBILD-Defrise phantom was used to evaluate cone-beam artifact reduction under a controlled setting (Figure 4). In the sagittal view, the ground truth contains 11 evenly spaced disks. FDK preserves only the central three disks, and SIRT recovers approximately the central five, leaving the remaining off-plane disks substantially blurred or lost because of cone-angle-related artifacts. VFINAF instead recovers all 11 disks with clear separation, indicating improved recovery of structures outside the central plane. This difference is most evident in the edge axial slice, which is the region most affected by cone-beam geometry. There, both FDK and SIRT

images show severe artifacts and inaccurate intensity in the edge disk, whereas VFINAF restores the FORBILD edge-disk structures with fewer visible artifacts and more faithful intensity patterns. In the central axial slice, where the data are less affected by cone-angle incompleteness, all three methods provide comparable image quality. The main advantage of VFINAF therefore lies in suppressing cone-beam artifacts and recovering off-plane structures that are degraded in conventional reconstructions.

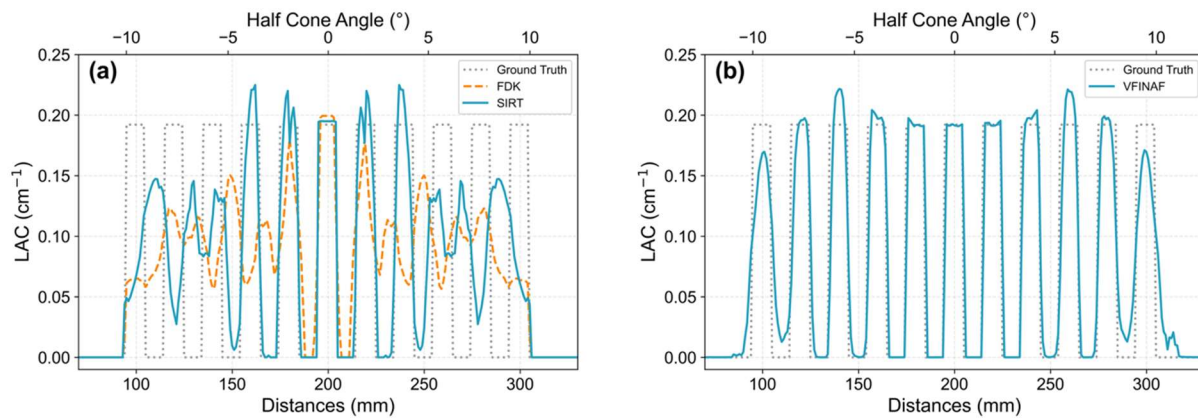


Figure 5. Line profiles of the linear attenuation coefficient (LAC, cm^{-1}) along the vertical direction in the FORBILD-DeFrise phantom. (a) Comparison among ground truth, FDK, and SIRT. (b) Comparison between ground truth and VFINAF. The upper x-axes indicate the corresponding half cone angles.

Vertical LAC line profiles through the same phantom were shown in Figure 5. FDK and SIRT both deviate substantially from the ground-truth profile, particularly at larger half cone angles, see Fig. 5(a). FDK shows strong intensity distortion and fails to preserve the expected disk-by-disk profile outside the central region (with $\pm 3^\circ$ half cone angle). SIRT recovers more central disks (up to $\pm 5^\circ$ half cone angle), but still exhibits pronounced blurring, peak distortion, and nonzero background between disks, indicating incomplete separation of off-plane structures. In contrast, the VFINAF profile in Fig. 5(b) closely follows the ground truth across the full vertical range and recovers all 11 disk peaks (with $\pm 10^\circ$ half cone angle) with improved separation and intensity fidelity. Only mild deviations remain near the largest half cone angles. These line-

profile results support the visual findings in Fig. 4 and indicate that VFANAF more effectively suppresses cone-beam artifacts while preserving the attenuation structure of the Defrise phantom.

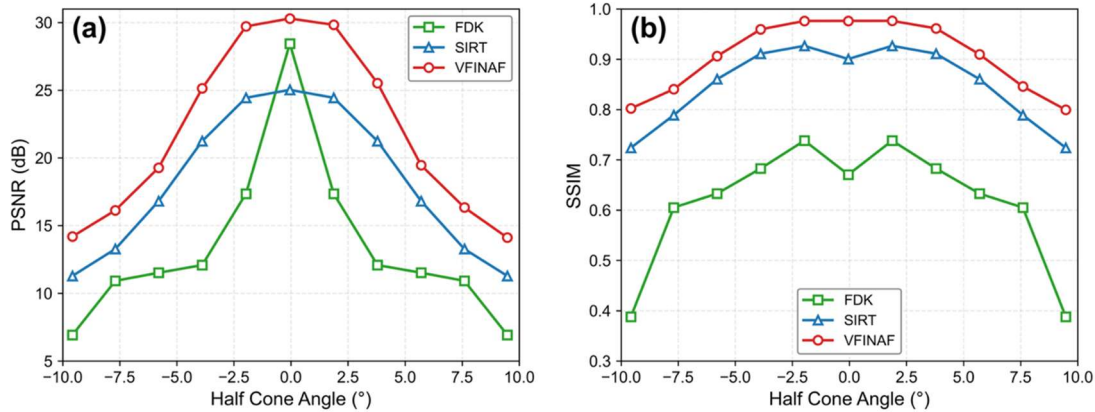


Figure 6. (a) PSNR and (b) SSIM as a function of half cone angle for the FORBILD-Defrise phantom; each value is the mean over the corresponding disk. Curves are shown for FDK, SIRT, VFANAF.

Figure 6 summarizes the PSNR and SSIM of each disk as a function of half cone angle. Both metrics varied systematically with half cone angle, with lower values toward the two peripheral ends of the disk array, where cone-beam artifacts were most pronounced. FDK showed the strongest angular dependence: its PSNR dropped rapidly to one-third of the PSNR at central disk, and its SSIM remained substantially lower than the other methods across most disks. SIRT produced a smoother profile and improved the peripheral disks compared with FDK, although its PSNR and SSIM still decreased noticeably away from the central region. VFANAF achieved the highest PSNR and SSIM across all half cone angles. In particular, its SSIM remained above 0.80 even at the peripheral disks and approached a high value near the center, indicating improved structural preservation over the full disk array. These quantitative results support the visual findings in Figs. 4 and 5, showing that VFANAF provides stronger cone-beam artifact reduction while maintaining high image fidelity across both central and off-plane regions.

3.B Experimental Results

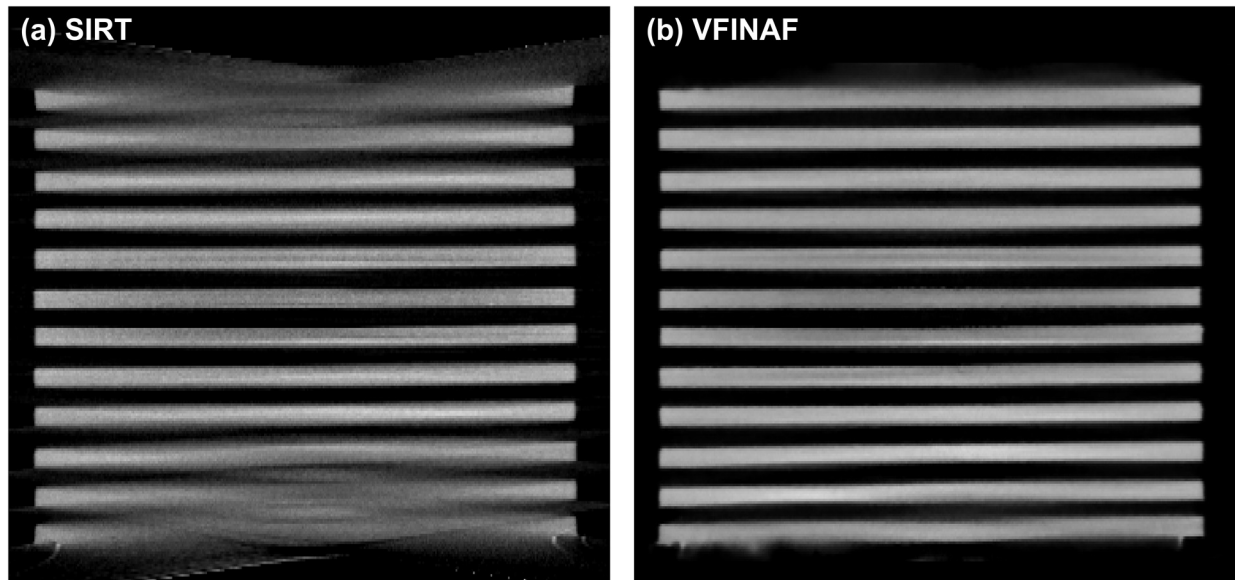


Figure 7. Sagittal views of the Defrise phantom reconstructed using (a) SIRT and (b) VFINAF at a cone angle of 10° . Window: $[-1000, 500]$ HU.

A similar reduction in cone-beam artifacts was observed in CT images of the physical Defrise phantom (Figure 7). In the sagittal view, SIRT preserves the expected repeated disk pattern, but shading and reduced disk-to-gap separation are evident near the two longitudinal ends of the phantom. As a result, only five disks can be clearly resolved in Fig. 7(a). VFINAF preserves the repeated disk structure with clearer separation between adjacent disks and resolves all 12 disks in Fig. 7(b).

The evaluation was also done using the anthropomorphic RANDO head phantom, with MDCT images acting as a reference (Figure 8). The MDCT images provide clearer depiction of the craniofacial bony anatomy in the axial, sagittal, and coronal planes. Compared with the reference images, the CBCT SIRT reconstruction images show pronounced degradation near both axial ends of the reconstructed volume, where cone-beam artifacts appear as intensity shading,

streaking, and loss of structural definition. VFINAF significantly reduces these artifacts across all three orthogonal views. The bony structures are better preserved than in SIRT and show closer visual agreement with the MDCT reference. These results indicate that VFINAF improves off-plane image quality in the RANDO phantom while preserving the main anatomical structures.

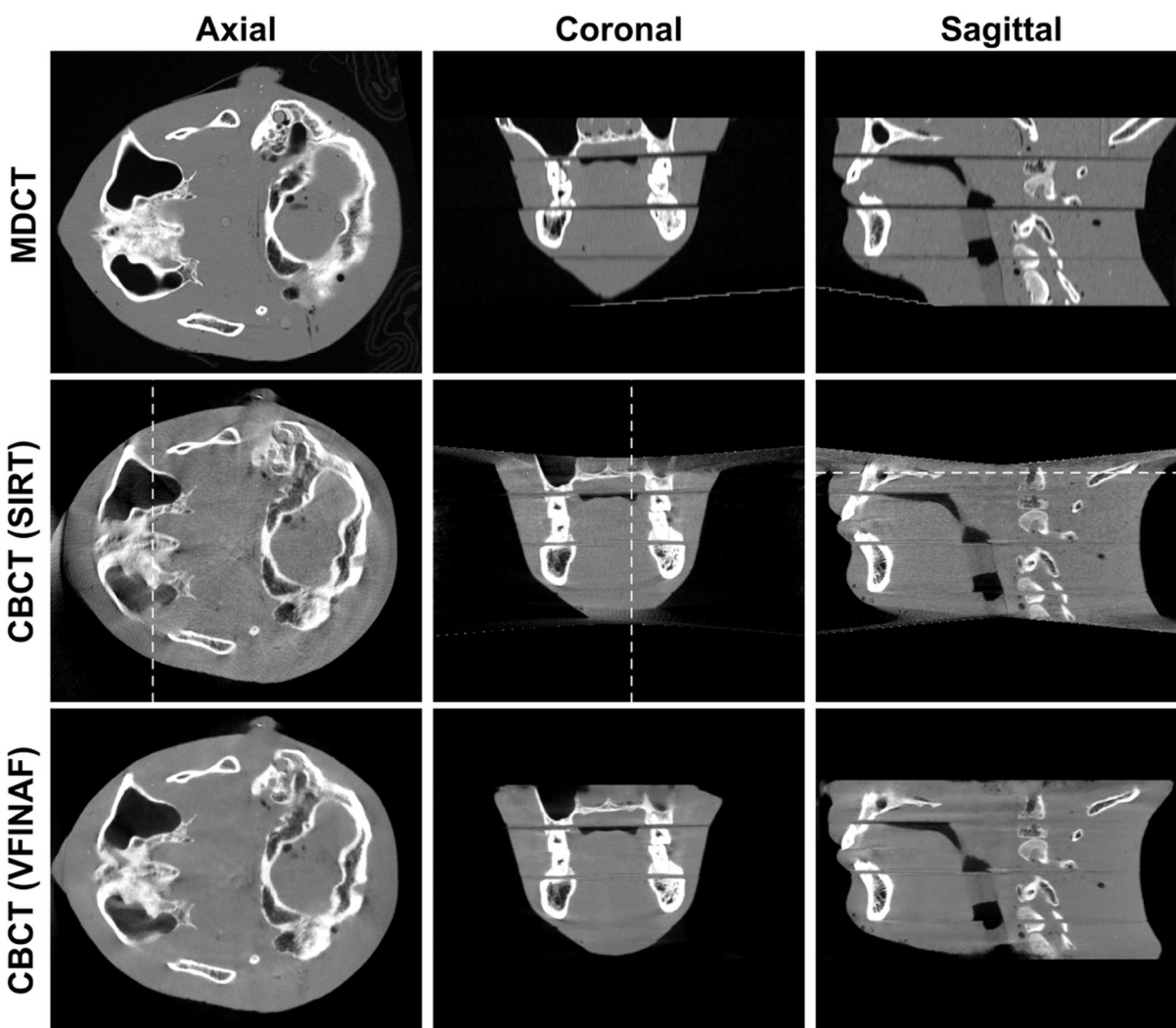


Figure 8. Orthogonal views of the RANDO head phantom reconstructed with SIRT and VFINAF, with MDCT shown as the reference. Axial, coronal, and sagittal slices are displayed for each image set. The dashed lines mark the slice positions used to generate the corresponding orthogonal views. Window: [-1000,1200] HU

311 3.C Effects of Re-projection

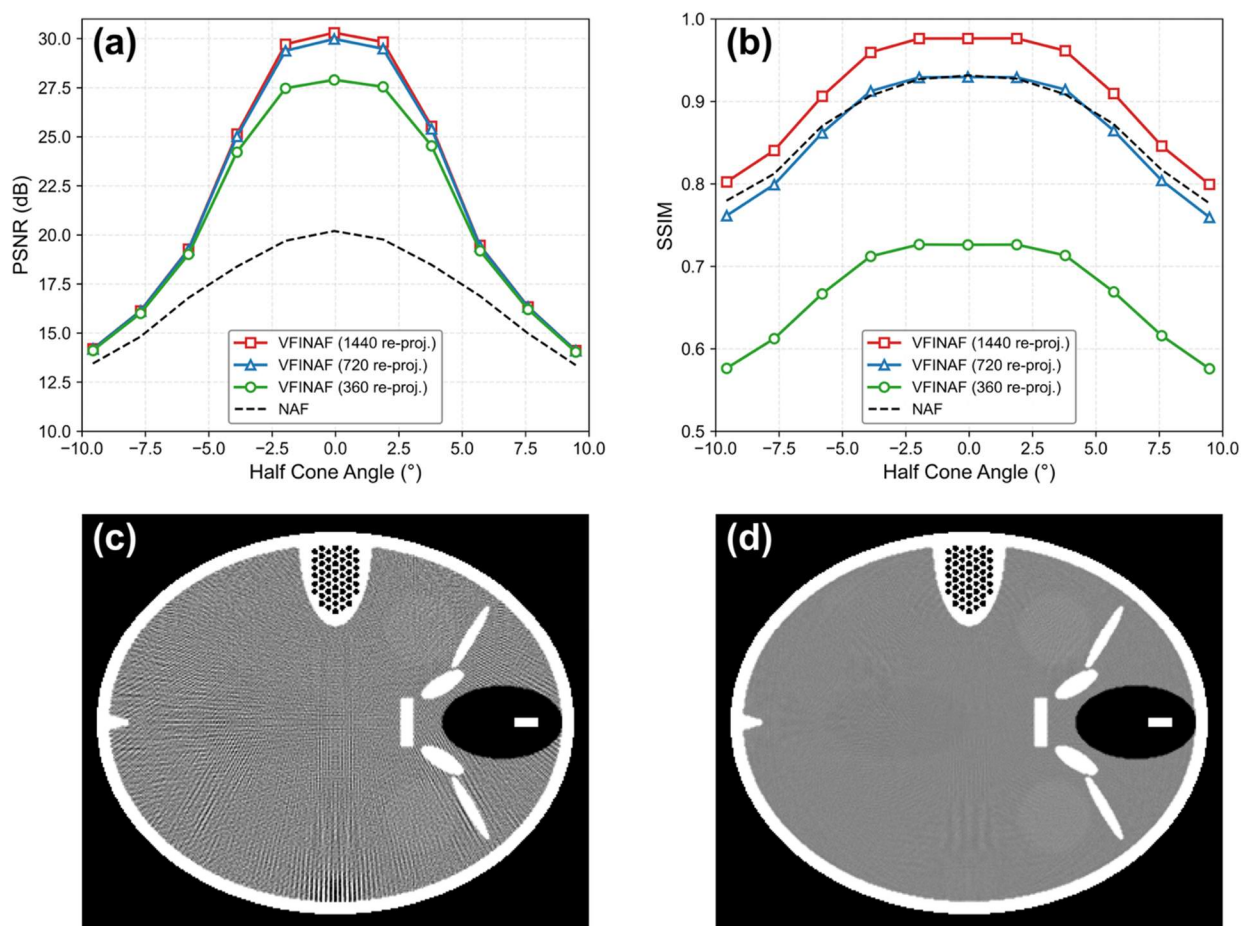


Figure 9. Effect of the number of fan-beam re-projection views on reconstruction quality in the FORBILD-Defrise digital phantom. (a) PSNR and (b) SSIM as functions of half cone angle. (c, d) VFANAF reconstructions using 360 and 1440 re-projection views, respectively. Window: $[0.15, 0.23] \text{ cm}^{-1}$.

Figure 9 compares direct voxel-wise evaluation of the trained NAF with VFANAF using different fan-beam re-projection views. Re-projection substantially improved PSNR across all half cone angles, with the improvement largely converging at 720 views (Fig. 9a). SSIM was more sensitive to angular sampling: 360 views produced residual view-aliasing streaks and markedly lower SSIM, whereas 1440 views consistently achieved the highest values (Figs. 9b). Further increasing the re-projection views does not increase the performance. This difference is also evident in the reconstructed images: pronounced residual view-aliasing streaks are present with

360 views but are substantially reduced with 1440 views (Figs. 9c and 9d). Based on these results, 1440 re-projection views were used throughout this study.

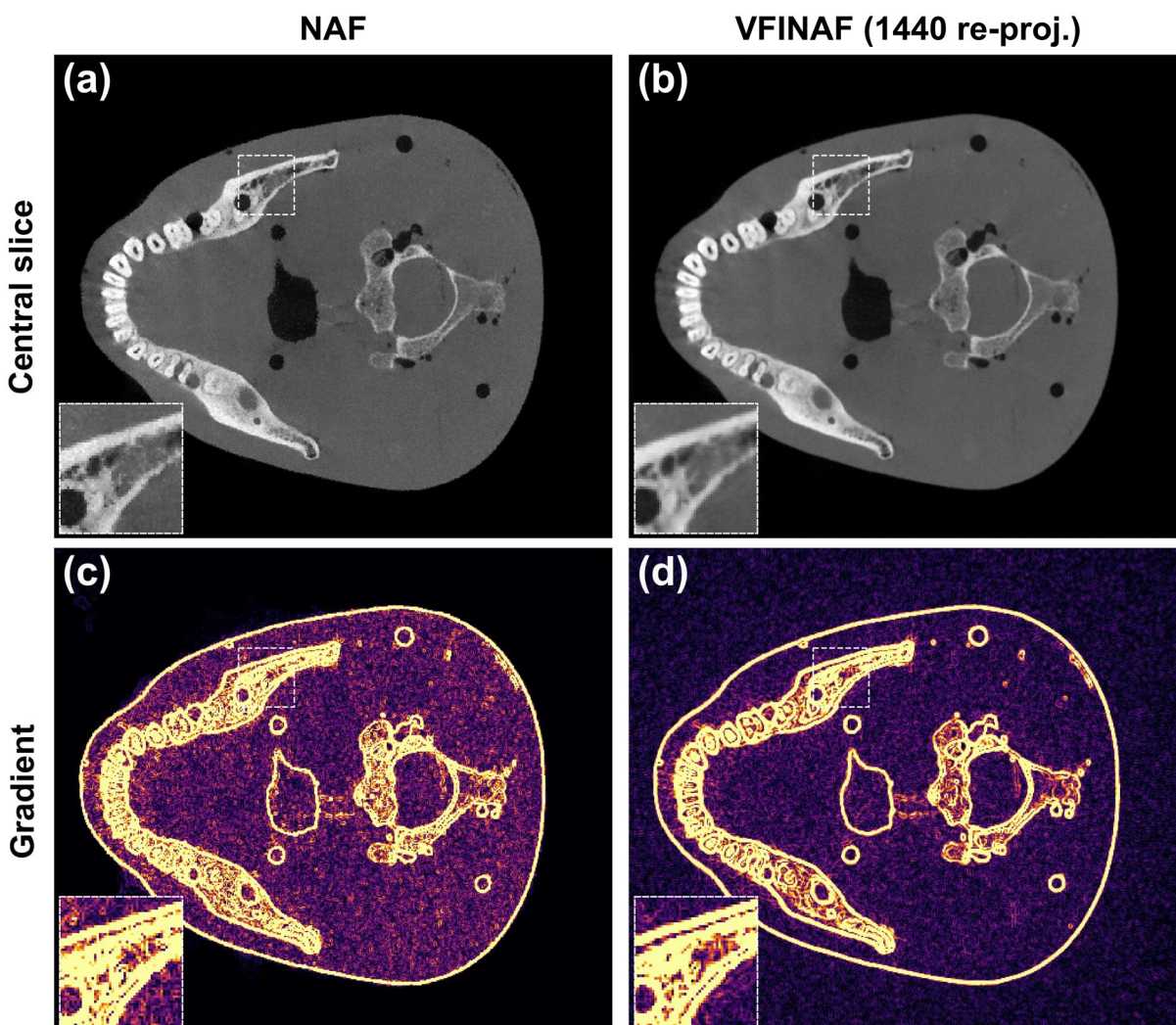


Figure 10. Central axial slice of the RANDO head phantom images using direct NAF readout (a) and VFANAF with 1440 fan-beam re-projection views (b). The bottom row shows the corresponding gradient-magnitude maps (c, d). Dashed boxes indicate the regions shown in the magnified insets. Window: $[-1000, 2000]$ HU.

Figure 10 evaluates the effect of dense fan-beam re-projection on the experimental RANDO phantom by comparing VFANAF using 1440 re-projection views with direct NAF readout for the central axial slice. Because an artifact-free reference was unavailable, gradient-magnitude maps were used to visualize noise-related intensity variations. Direct NAF readout exhibits visibly higher noise throughout the phantom, whereas VFANAF produces a more uniform reconstruction

with lower overall noise while preserving the anatomical structures. This difference is more apparent in the gradient maps: direct NAF readout shows widespread background responses, while VFINAF substantially suppresses these responses, leaving the dominant gradients concentrated around anatomical and material boundaries. These results demonstrate that dense fan-beam re-projection reduces noise relative to direct NAF readout in the experimental data. Together with the re-projection-view analysis in the digital phantom, this finding supports the use of 1440 re-projection views in the remaining experiments.

4. Discussion

This study demonstrates that VFINAF reduces cone-beam artifacts in CBCT by using a learned continuous neural attenuation field for dense virtual fan-beam re-projection. The resulting full-angle dense-view fan-beam sinograms support high-quality slice-wise 2D FBP or SIRT reconstruction, even though the experimental acquired projection data is in cone-beam geometry. In Figures 4–6, VFINAF and conventional reconstructions are comparable near the central plane, but their performances differ significantly at larger half cone angles: FDK preserves the structural detail only up to $\pm 3^\circ$ half cone angle, and SIRT up to $\pm 5^\circ$ half cone angle, while VFINAF recovers the structural detail up to $\pm 10^\circ$ half cone angle. The physical Defrise experiment shows the same conclusion as the digital phantom simulations. In the RANDO phantom experiment, VFINAF reduces cone-beam artifacts in all three orthogonal planes, with bony structures showing closer visual correspondence to the MDCT reference (Figure. 8). Figures 9 and 10 further show that dense fan-beam re-projection improves upon direct NAF readout by suppressing local fluctuations in the learned field through tomographic reconstruction. Adequate angular sampling is necessary, as 360 views introduce

aliasing streaks, whereas 1440 views provide the best structural fidelity and noise suppression. The greater residual variations in the edge slice indicate that re-projection reduces, but does not eliminate, uncertainties associated with Tuy's incompleteness. The distinguishing contribution of VFINAF is geometry-decoupled fan-beam re-projection. Unlike earlier methods that use the trained NAF as a direct image or synthesize dense-projections views in the same acquisition geometry, VFINAF utilized the learned NAF to synthesize dense view in a fan-beam geometry slice-by-slice, thereby decoupling from the original cone-beam acquisition. This enables significant cone-beam artifact reduction while preserving the unsupervised advantages of NAF. Future work will evaluate VFINAF on clinical patient images to test whether the observed off-plane artifact reduction generalizes to other imaging acquisition geometry such as multiple source CBCT.

5. Conclusions

VFINAF provides an unsupervised NAF-based reconstruction framework for reducing cone-beam artifacts in CBCT. By learning a NAF from cone-beam projections and reconstructing it through dense fan-beam re-projection, VFINAF decouples the final reconstruction geometry from the original acquisition geometry. This allows each axial slice to be reconstructed from a complete fan-beam sinogram and avoids Tuy's incompleteness associated with direct cone-beam reconstruction.

In both the digital phantom simulation and real-data CBCT experiments, VFINAF reduced cone-beam artifacts in all three orthogonal planes. Dense fan-beam re-projection also reduced noise compared with direct NAF readout when sufficient angular sampling was used. These findings

suggest that geometry-decoupled fan-beam re-projection is the key contribution of VFINAF, enabling cone-beam artifact reduction while retaining the unsupervised advantages of NAF reconstruction. Future work will evaluate whether these benefits generalize to sparse-view and limited angle CT acquisition geometry.

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